// Genova Mongalo

(510) 274-1272 | genova@gmango.dev | linkedin.com/in/GMongalo

Education

University of Missouri-Kansas City(UMKC) Kansas City, MO
Bachelor of Science in Computer Science, summa cum laude *Aug. 2021 – Dec 2024*

Georgia Tech Atlanta, GA
Masters in Computer Science - Emphasis on Artificial Intelligence *Expected 2027*

Experience

Founding AI/ML Product Engineer September 2025 – Present
Stealth Startup *Kansas City, MO*

- Ported clinically validated mobility assessments from MATLAB research prototypes to production Swift, matching the clinical reference within 0.56% on a 15-biomarker postural-sway test and 10-meter-walk gait speed exactly.
- Built the on-device biomarker pipeline in Swift: 100 Hz CoreMotion capture, gravity removal, zero-phase Butterworth filtering, double integration, and PCA into 15 biomarkers with automated quality flags, computed locally so raw patient motion never leaves the device.
- Built a speech assessment pipeline pairing on-device Apple Speech for live partials with a Whisper backend and LLM transcript normalization, surfaced through a patient-review panel to capture the communication deficits affecting one in three stroke survivors.
- Implemented a context-aware clinical assistant with patient-context retrieval and PubMed/NCBI evidence via Entrez, running on self-hosted inference with Flask REST and WebSocket streaming under 1s latency, keeping protected health information on infrastructure we control.

Federal AI/ML Engineer May 2025 – August 2025
AFRL Sensors Directorate Contracting Program, University of Dayton *Dayton, OH*

- Pioneered the development of generative AI models enabling a CNN classifier to successfully recognize real objects when trained on generated data when utilized in transfer learning framework, addressing AFRL/DoD priorities.
- Using PyTorch, adapted state of the art classifiers to handle novel imaging modalities.
- Created an innovative game theory approach to GAN training, significantly improving the results of the generator's development, trained at scale using Slurm-parallelized compute.
- The generative model's output led to an increase of 50 percentage points in the downstream classifier compared to the baseline.

AI ML Engineer Intern June 2024 – Dec 2024
NSF REU AI-Empowered Cybersecurity - University of Missouri, Kansas City *Kansas City, MO*

- Engineered an LLM to detect ransomware in Industrial Control Systems, achieving 99%/91% binary/family classification accuracy; presented at IEEE Big Data 2024 (solo author, mentor-guided).

Presentation / Leadership

Mongalo, G. (Dec. 2024). LLM Based Approach to Real Time Ransomware Detection for Industrial Control Systems. Presented at IEEE Big Data 2024, Washington, D.C. (Presenter & Author, solo work with mentor guidance).

Projects

AWS DeepRacer | *Reinforcement Learning, AWS*

- Trained and tuned reinforcement-learning policies for a 1/18-scale autonomous vehicle on AWS, iterating on reward functions and evaluating driving performance: hands-on autonomous control and cloud-based model training.

Technical Skills

Languages: Python, C / C++, Swift, SQL (Postgres), Java / C#, JavaScript
Frameworks: scikit-learn, Flask, SwiftUI, Socket.IO, FastAPI
Developer Tools: Git, Xcode, AWS, Google Cloud Platform, VS Code, Anaconda
Libraries: pandas, NumPy, Matplotlib, PyTorch